

Missing Values Visualization Challenge for Airbnb Athens Listings

Motivation

Understanding missing data is a prerequisite for reliable modeling and feature engineering. In the Airbnb dataset, null values may signal either incomplete entry or meaningful structural absence (e.g., shared rooms lacking bedroom counts).

The goal is to identify:

- which variables have the largest proportion of missing entries
- whether missing data clusters within certain listing types
- if multiple columns tend to be missing together (co-missingness)

Key ideas

We implemented four complementary visual analyses using PySpark for aggregation and Matplotlib for visualization:

- 1. Column-wise Missingness**
 - a. Bar chart ranking columns by percentage of missing values
 - b. Helps prioritize cleaning or imputation focus (e.g., `reviews_per_month`, `review_scores_rating` most affected)
- 2. Row-wise Missingness Distribution**
 - a. Histogram showing how many nulls each record contains
 - b. Indicates the proportion of fully complete listings and whether row deletion is feasible
- 3. Missingness by Group**
 - a. Heatmap of %null per column within each `room_type`
 - b. Reveals structural missingness (e.g., some fields only missing for certain room categories)
- 4. Pairwise Co-missingness (Jaccard Index)**
 - a. Correlation-style heatmap measuring how often two columns are missing together
 - b. Detects coupled gaps (e.g., listings without reviews usually also lack review scores)

Limitations

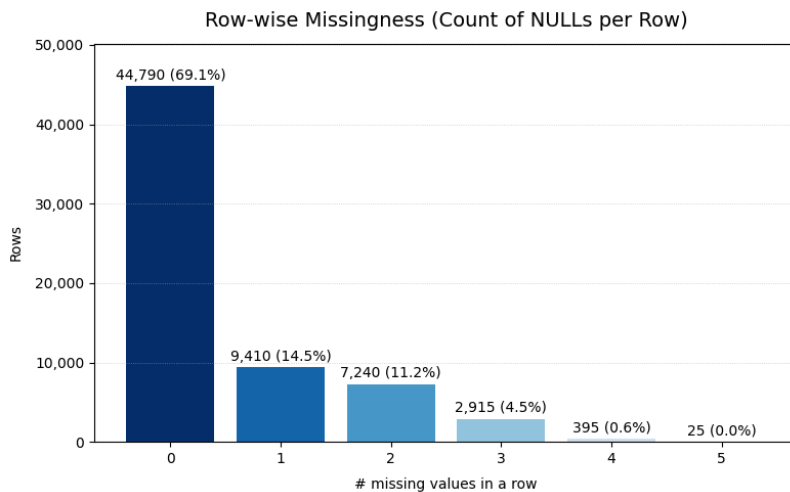
- **Scalability** — pairwise Jaccard computation scales quadratically with the number of columns; we apply it only to a selected subset.
- **Assumptions** — blank strings were treated as null; this assumes blanks are genuine missing entries, not valid zero-length values.

Design Decisions

- **Color Palette:** sequential blues and purples for consistency; darker shades denote higher missingness.
- **Labels and layout:** dynamic label wrapping, right-aligned text for long column names, and auto-adjusted figure sizing to prevent overlap.
- **Interactivity:** static Matplotlib chosen for reproducibility; code can be extended to Plotly or Databricks dashboards if needed.
- **Styling consistency:** all plots share unified typography, grid style, and title formatting for cohesive presentation.

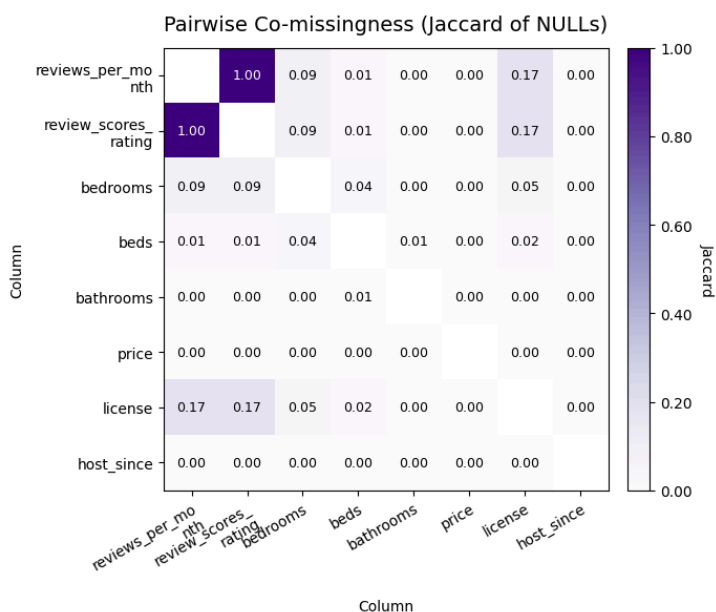
Sample visualizations

1. Figure 1: Row-wise missingness distribution



This histogram displays the count of missing values per record. The chart shows that about **69%** of listings are fully complete (**0 missing values**), and most others miss only **1–2 attributes**.

2. Figure 2: Pairwise co-missingness



This heatmap reveals how strongly certain variables share missing patterns. The darkest cells (**Jaccard ≈ 1.0**) indicate that **reviews_per_month** and **review_scores_rating** are almost always missing together — confirming a clear dependency between review-related attributes.